Detailed research plan (5000 words) – currently 5034

What is the problem being addressed?

Dizziness

Dizziness ranks among the top reasons for visiting a doctor, and affects up to 35% of individuals at some point in their lives. It can be due to neurological, inner ear (vestibular), cardiovascular, psychological or other disorders.

A German study provides the most cited population wide estimate of the incidence and prevalence of vertigo caused by vestibular conditions (Neuhauser et al 2005). They reported vertigo was caused by vestibular conditions in 24% of cases, had a lifetime prevalence of 7.4% between the ages of 18-79 years old, and a 1-year incidence of 1.4%.

What is the vestibular system?

The vestibular system incorporates the inner ear (vestibular apparatus), vestibular nuclei in the brainstem, the eighth cranial nerve and the cerebellum. It is responsible for maintaining clarity of vision and contributing to our control of upright balance and movement.

The vestibular apparatus is located in the temporal bone on both sides of the head. It consists of a bony and membranous labyrinth, in turn filled with perilymph and endolymph fluid. The membranous labyrinth is formed of three semicircular canals and two otolith areas. They detect angular and linear accelerations respectively, as well as the otoliths also detecting head position. The receptor areas are characterised by hair cells covered by a gelatinous membrane, with each hair cell connected to the eighth cranial nerve. The receptor areas differ in that the otoliths have calcium carbonate crystals embedded on the receptor gelatinous membrane.

Damage to any part of the vestibular system can result in dizziness, vertigo, blurred vision, imbalance and falls. Symptoms can vary and may be spontaneous, triggered by movement or environments, the result of exposure or trauma and can also be persistent.

Positional dizziness

Positional dizziness is triggered by changes in head position relative to gravity's influence. For example, getting into bed, rolling in bed and bending over. It can result in dizziness, vertigo, and blurred vision, with unsteadiness commonly reported. The symptoms are typically brief, but can have a significant impact on a persons independence in everyday activities (DHI Ref no 4 Dan), mood, walking, confidence and can cause falls.

Positional dizziness can be due to inner ear or brain disorders. Gold standard positioning manaourvres should be conducted to help make a diagnosis (Ref). However, these tests rely on correct interpretation of the induced eye movement (nystagmus) and this can be challenging especially for someone new to assessing dizziness (Male et al ...; Meldrum et al ...).

It is important to identify the underlying disorder, as the treatment and management will be very different. Repositioning manaourvres successfully treat and resolve BPPV, whereas in central disorders, these do not work and medication is typically required alongside treatment to address any underlying structural cause.

Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients/service users, carers, and health and care services?

This research has been developed following patient and service user feedback regarding the challenges in assessing, identifying and treatment positional dizziness.

A 35 year old black female recently reported "*...I am frustrated that I have had to suffer with this horrible dizziness for so long. Why couldn't I see someone to sort it out quicker as this has stopped me living my life?...*".

Additionally, a 66 year old white female reported "*...everyone kept asking me if I was dizzy, and I kept telling them no. But I was really unsteady getting out of bed, and when I looked up. My doctor referred me for physio thinking it was my balance due to my hip and back pain. It wasn't until you did the tests that you found the real cause was in my ears and cured me in a matter of minutes...*".

These two quotes highlight some of the common complaints voiced by patients I have worked with over my clinical career. Accessing effective assessment and treatment of a peripheral condition, such as BPPV, can reduce the risk of falls and injuries (Ref), allow someone to independently live their lives, and reduce the burden on the healthcare and social care services. In the elderly, falls can lead to death, wither as a result of the fall or indirectly (Ref 5 Dan). Whilst no cost analysis has been presented in the research, enabling people to remain active is crucial for general health and wellbeing and is advocated by both the NHS (Ref) and World Health Organisation Physical Activity Guidelines (Ref).

Another patient that raised concerns and a desire for change was a doctor that came to see me. She was a 32 year old female who had been self-treating her positional dizziness for several weeks, but was vomiting afterwards and the issue was not resolving. Whilst her nystagmus appeared to consistent with a peripheral cause, the lack of resolution and vomiting concerned me (MacDonald...). In collaboration with a medical colleague, and after a through examination to include a clear brain scan, a central positional diagnosis was made. Specifically, vestibular migraine. My colleague started her on migraine prophylaxis and her dizziness completely resolved. She reported "*...I did not know that I had migraine as I do not have a headache, and thought only BPPV caused this type of dizziness...A better awareness of this problem is needed*". The stereotype that migraine means bad headache, unfortunately means it frequently goes un-diagnosed.

I have been involved in training GPs and other medical professionals, in the assessment of dizziness, and more specifically in recognising BPPV. At these events, GP's have told me "*...they do not have time...*", "*...prefer to signpost patients to someone else to manage the problem...*", and "*...do not feel confident in interpreting the eye movements...*". This latter point was also seen in a recent acute vertigo study, whereby doctors in the accident and emergency department made diagnoses of BPPV without doing the required tests, and diagnosed BPPV when a proportion of them actually had a central diagnosis instead (Ref).

This research aims to simplify the diagnosis of peripheral versus central causes of positional dizziness and unsteadiness, and thus reduce the time patients may be waiting to receive the correct treatment, whether that be for a peripheral of central cause of positional dizziness.

Review of the existing evidence - how does the existing literature support this proposal?

Benign Paroxysmal Positional Vertigo

The most common inner ear disorder, worldwide, is benign paroxysmal positional vertigo (BPPV). In the previously mentioned German population interview (Ref), vertigo was due to BPPV in 8%, had a lifetime prevalence of 2.4%, and a 1-year incidence of 0.6%. Furthermore, they describe that 1 in 3 of participants who reported vertigo due to a vestibular disorder, were diagnosed with BPPV.

Relying on the history taking alone, may fail to identify this treatable cause of dizziness or unsteadiness. This was demonstrated in adults over the age of 60 years old, who were diagnosed with BPPV, but did not report vertigo or dizziness, and instead conveyed unsteadiness, imbalance and light-headedness (Li et al 2024; Batuecas-Caletrio et al 2013).

Additionally, vestibular agnosia has been identified in older adults, those with neurodegeneration and following traumatic brain injury (Smith et al ...). Here patients had a positive positional test suggesting BPPV, that resolved with repositioning manaourvres, but the patients did not perceive any illusion of movement resulting from the observed nystagmus.

These studies emphasise why positional tests should be completed in order to make a diagnosis. Furthermore, as BPPV can cause falls, or be the result of a fall, the World Falls Guidelines (Ref) have also strongly recommended that patients who have fallen, should have a multifactorial falls risk assessment completed. They specify this includes a vestibular assessment and positional tests, in order to conduct and manage any modifiable risk factors.

Diagnostic criteria have been published by the Barany Society (Von Brevern et al 2015), that state the clinical features essential during canal specific positioning manoeuvres, to include nystagmus latency, direction, and duration.

Each of the three canals, in either inner ear, can be affected by BPPV to different extents. The most commonly affected is the posterior canal accounting for 85-95% of cases (Parnes et al 2003) and horizontal canal for 5-15% (Cakir et al 2006; White et al 2005). Anterior canal involvement is reported as only 1%, with a central cause strongly suspected instead of a peripheral cause (Korres et al 2002).

There are certain key treatment manaourvres, that are used to treat each canal that can be affected by BPPV, for example the Epley manoeuvre. These treatment manaourvres move any displaced otoconial debris out of the semicircular canals, by moving the patients head and body, through a series of specific positions. Each specific position is designed to put the affected canal under the maximal influence of gravity.

Resolution rates for the most common type of BPPV, after one treatment are 72-74% (Ref), and increase to 90% after two treatments (Ref), depending on the manoeuvre selected. However, there is limited evidence on the resolution rates of all the common sub types of BPPV, and there is also lack of consensus on the number of treatments required to successfully resolve all types of BPPV. This was highlighted in a survey I completed in 2024 to a group of 24 multidisciplinary professionals, attending a specialist conference for dizziness in the UK.

BPPV can recur at a rate of xxx per year (Ref). Certain cardiovascular and metabolic risk factors have been proposed, however there are differing outcomes in the literature and methodological concerns in how these studies have been undertaken. Of interest, no studies to date, have

explored if there is an anatomical variation in the inner ear, specifically the canal orientation, that may explain why the specific movements of the treatment procedures do not work. This is one of the aims of this proposed study.

Central positional nystagmus

Brain disorders that can cause positional symptoms include damage to the cerebellum or brainstem from stroke, tumours or more commonly due to non-structural causes such as vestibular migraine (Lemos...).

Whilst specific nystagmus features can help identify positional dizziness from the brain, at present the diagnosis is often one made due to failure of an inner ear cause (BPPV) to resolve (Ref). Thus patients may need to attend multiple appointments in the health service prior to a central cause being considered. Or, they may continue to have treatments, and a central cause is never suspected.

However, there are gaps in the literature and a lack of consensus in how many treatments are deemed adequate in order to resolve BPPV. Therefore relying on "failure to respond" as a diagnostic feature, when there lacks certainty how many treatments should resolve BPPV, may contribute to a lack of identification of central causes.

Central positional nystagmus (CPN) is also suspected when atypical forms of nystagmus are seen. Yet, emerging research now suggests there may be atypical forms of BPPV that may account for these nystagmus patterns, complicating the process of diagnosis, and increasing the risk that brain causes may be overlooked and not managed.

Whilst the treatment manoeuvres for BPPV are safe (refs), the treatment manaourvres will not resolve a central cause of positional nystagmus and dizziness. Furthermore, the resulting delay in identifying that the cause is central, may lead to increased levels of anxiety, low mood, social withdrawal, increased dependence on others and potentially falls.

When a central cause is identified, it is important to address any structural abnormality found on a brain scan. This may involve certain medications or surgical intervention. However, the most common central cause is vestibular migraine, and would not show any abnormality on a brain scan. In this instance, migraine prophylaxis is recommended (Ref 35 Lemos). This was the course of action taken by Mahrous and colleagues (Ref). They treated ten patients from a case group of 12 with a specific medication used in migraine, and the positional symptoms completely resolved. However, they only did this after treating them with BPPV manoeuvres three times, assuming the positional symptoms had to be due to BPPV first.

Whilst CPN is recognised as a cause of positional nystagmus, there is a lack of awareness in making the diagnosis both within the research literature and thus clinically. In a systematic review of all the literature to date, Lemos and colleagues (Ref) recommend further research using video goggles, whilst conducting standardised assessments using the testing manoeuvres commonly used in BPPV. They also recommend longitudinal studies to evaluate patients over time, and to understand the pathology in greater detail.

What are the research question/aims and objectives?

This study aims to prospectively assess patients presenting with dizziness and/or unsteadiness that is triggered by head and body position changes, in order to:

Primary aim

- characterise the nystagmus features and establish accompanying symptoms for peripheral and central causes positional causes

Secondary aims

- determine the prevalence of peripheral and central causes of positional nystagmus in the North Central London area
- identify the number of treatments required for each canal type of BPPV within the cohort and through an International modified Delphi Consensus
- establish the recurrence rates for different types of BPPV
- explore if there is an anatomical difference to the inner ear for those with non resolving BPPV

This study aims to expand the limited literature to help clinicians identify and distinguish between peripheral and central positional nystagmus more effectively. This will improve access to the correct treatment in a more timely manner, and enhance patient care and experience.

The study will also fill a knowledge gap in understanding how many treatments are required for each sub type of BPPV. Prior to this application, I conducted a preliminary survey to explore the overall management of BPPV across a group of 24 multidisciplinary professionals, attending a specialist conference for dizziness in the UK. The results of this have informed the research protocol, but a larger scale Delphi consensus opinion will be sought as part of this proposal to inform best practice of BPPV, worldwide.

Finally, this study will include a novel case control exploration as to whether there is an anatomical explanation for why BPPV does not respond to the gold standard manaourvres. This will be the first study, to the applicant's knowledge, using 3D reconstructive magnetic resonance imaging in peripheral positional nystagmus.

Project Plan

Observational study

Recruitment

Adults (over 16 years old) reporting dizziness, vertigo, or unsteadiness that is triggered by head or body movement will be eligible to participate. Exclusion criteria include: medical instability (cardiovascular, cervical, or neurological), spontaneous nystagmus in sitting, pregnancy, previous history of existing and persistent vertigo, and lacking capacity to consent. Those unable to understand English will also be excluded, as the questionnaires used in this study and in the vestibular specialism, have not been validated and translated into every language.

This prospective study will be open for recruitment for 18 months. A purposive sample will be recruited. Figure 1 summarises the flow of each participant through the study duration.

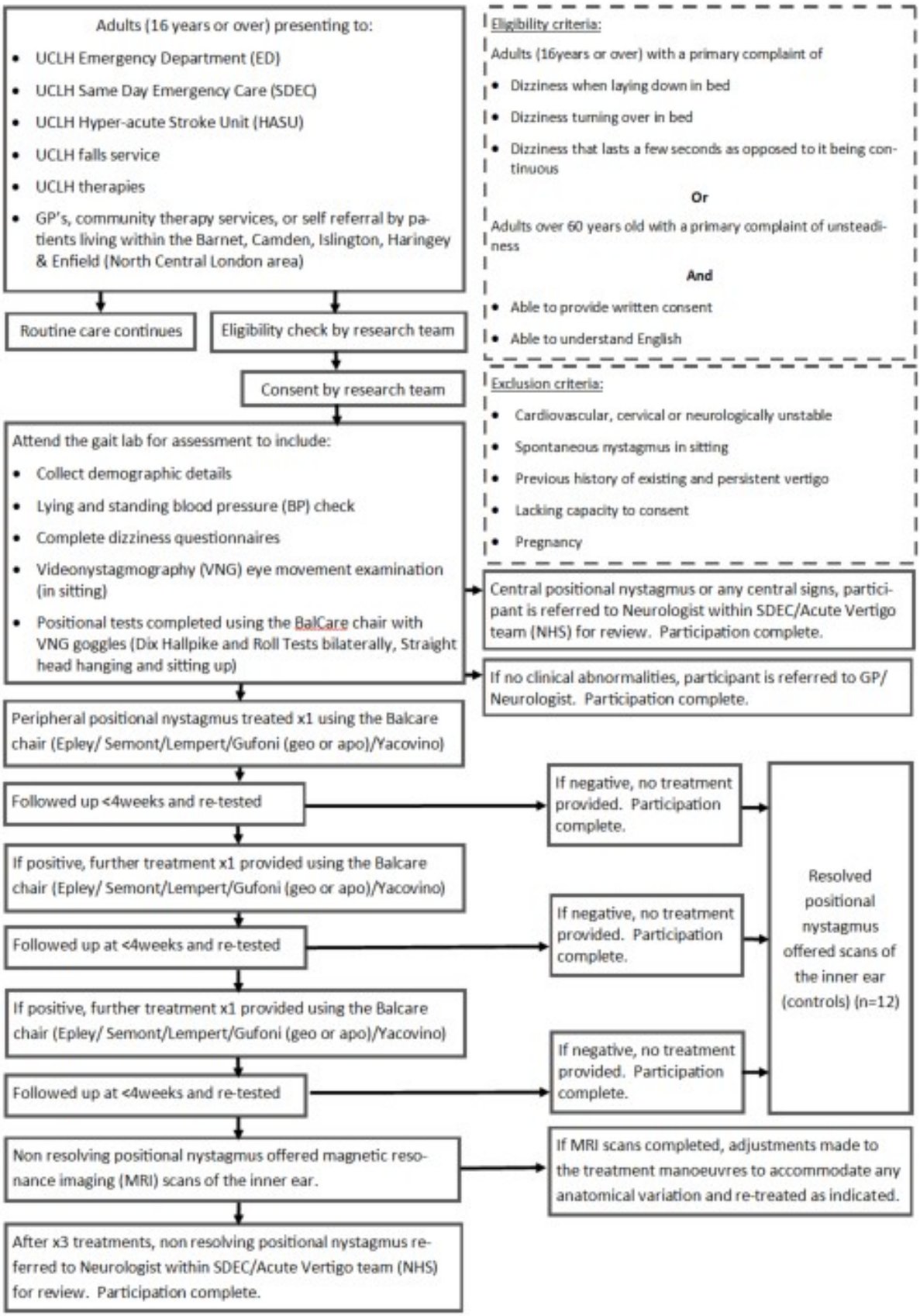

Figure 1: Participant flow through the study.

Power Calculations

Current research lacks detailed prevalence data on positional nystagmus across healthcare settings, failing to differentiate between peripheral and central causes. The most relevant data from a French neuro-otology clinic over nine months indicates that 20% of patients with symptoms of vertigo and unsteadiness showed positional nystagmus, with breakdowns into BPPV (80%), other peripheral causes (3%), central causes (12%), and undetermined (5%) (Bertholon *et al.*, 2006).

To ensure sufficient statistical power for detecting the 20% prevalence rate of positional nystagmus identified in previous studies (Bertholon *et al.*, 2006), we calculated the necessary sample size using the formula $n= [(z)^2 (p)(1-p)]/(d)^2$ where Z is 1.96 for 95% confidence, p is 0.20, and d is 0.05 (Sharma *et al.*, 2019; Pourhoseingholi *et al.*, 2013). This calculation resulted in a sample size of approximately 246 adults, which is sufficient to provide reliable data while ensuring the study remains scalable and manageable within the constraints of our resources.

Participant recruitment

The study will be conducted at a single site, based at The National Hospital for Neurology and Neurosurgery. This will ensure access to specialist equipment (Balcare chair) for standardisation of testing, to address the study aims and as recommended by researchers (Lemos). The study will be embedded within the Acute vertigo service, and Same Day Emergency Care service (SDEC), at University College London Hospitals (UCLH) with oversight by Consultant Neurologists. This would also be the location patients from the North Central London area would attend for routine clinical care, and hence this reduces any additional travel burden or cost.

The North Central London (NCL) area is home to a diverse range of ethnicities, that includes white, Asian, and Black people. The largest ethnic minority communities in this area of London are Turkish, Irish, Polish and Asian. Discussions with patients from the NCL area, revealed they wanted to be involved in this study, but raised concerns about covering the travel costs. One person (Mr M) also raised concerns about other people he knew being able to fully understand and speak English well enough to take part. Therefore, funding will be requested to cover travel expenses and also cover costs of any interpreters that are required.

The study will be advertised through local GP network and community services for North Central London. This will include approximately 200 GP surgeries, and community neurology teams in Barnet, Camden, Enfield, Haringey, and Islington. It will also be advertised internally at UCLH through email communication and attendance at departmental meetings. This broad recruitment strategy will ensure the desired sample size will be met in the allocated recruitment time.

Advertisements will include screening questions for positional dizziness and unsteadiness. These questions have been adopted from a diagnostic framework for vestibular causes of dizziness (Male et al), which had PPI involvement and a modified Delphi consensus.

If potential participants answer the questions as indicated on the advertisement, the participant or clinician will be encouraged to contact the research team to make a referral. Patients helped to determine the best recruitment approach, and reported they would like an email and a telephone option. These contact details of the research team will be included on the advertisements.

Participants will be able to self-refer, or be referred by their GP or community therapy services from within North Central London areas. Internal referrals from UCLH emergency department (ED), falls service, the hyper-acute stroke unit (HASU), and from therapy services will also be accepted.

Potential participants will be contacted by a member of the research team and provided with the participant information sheet by email. They will be given the opportunity to ask questions, prior to providing consent. Consent will be provided electronically through a central REDCap database set up for this project. If potential participants request the consent form in paper format, the signed hard copy will be loaded in to the secure database.

They will then be invited to attend the hospital for assessment and treatment as the earliest point in time. The GP will be informed of their participation in the study.

Data Management

Potential participants' data will be identifiable for the purpose of follow up, and potential 3D scanning. However, it will be anonymised for any dissemination. All data will be kept confidential and secure by the research team and stored electronically in a central REDCap database. The REDCap database will require a password to access which will only be known by the research team.

Eye movement recordings will be stored on the Balcare laptop and secure drive. This will require a password to access, and the data will be handled securely in line with UCLH routine data governance regulations.

Research protocol

Participants will electronically complete routine questionnaires used in dizziness and balance services (see Table 1) and provide more personal information relating to their health. This will help identify any accompanying symptoms for peripheral and central causes of positional nystagmus.

Patient reported questionnaire	Details
Dizziness handicap inventory (DHI)	A measure of self-perceived disability attributable to vestibular disease. Subjects score 25 questions which are classified into three domain levels; Yes = 4, Sometimes = 2, No = 0 with a total percentage score calculated. Scores >20 indicate dizziness/unsteadiness is having a significant impact on their life (Jacobson and Newman, 1990).
Situational Vertigo Questionnaire (SVQ)	Evaluates the extent that certain situations cause symptoms of vertigo or makes patients feel worse. Each activity rated between 0 (not at all) and 4 (very much) with a not tried (NT) option available. Scores >1 indicate visual induced dizziness and/or imbalance (Guerraz *et al.*, 2001).
State-Trait Anxiety Inventory (STAI – form Y)	A self-reported assessment of trait and state anxiety.

	Subjects rate 20 statements using a 4-point Likert scale between 1 (not at all) to 4 (very much so). Higher scores indicate greater anxiety (Julian, 2011).
Visually Induced Motion Sickness Susceptibility Questionnaire (VIMSSQ - short)	A self reported tool to estimate the susceptibility to motion sickness when using devices and watching moving visual content. Subjects rate six devices for how often they experience nausea, dizziness, headache, fatigue, and eye strain using a 4-point Likert scale. Higher scores indicate greater motion sickness susceptibility (Golding *et al* 2021)

Table 1: Details of the questionnaires each participant will complete as part of the initial visit.

Participants will then be assessed, by a specialist physiotherapist, using a purpose-built chair (Balcare). This chair allows accurate assessment, irrespective of the patient's physical mobility level or co-morbidities, and without any increased manual handling demands for the clinician. It provides a standardised approach to assessment and uses video goggles that helps clinicians to interpret different types of nystagmus. This will allow more accurate interpretation and quantification of peripheral positional nystagmus from central positional nystagmus.

Each assessment visit will take no more than one hour. It will include a lying and standing blood pressure check for any postural hypotension, and a full eye movement examination in sitting using the video goggles (fixation removed). The eye movement examination includes observing for spontaneous nystagmus, gaze evoked nystagmus, and abnormalities in smooth pursuit and saccades. Positional tests using the Balcare chair will then be completed, whilst wearing the video goggles. These will include Dix Hallpike on both sides, Roll Tests, straight head hanging and sitting back up.

Should BPPV be identified as per the diagnostic criteria (ref), then the appropriate treatment manoeuvre will be provided once using the Balcare chair. This may include the modified Epley manoeuvre, Semont liberatory manoeuvre, Gufoni manoeuvre (geotropic or apogeotropic), Lempert BBQ roll, or the Yacovino manoeuvre (Refs?). Each of these manoeuvres take no more than 3 minutes to complete using the Balcare chair.

These patients will be followed up at the next convenient time (within four weeks) and re-treated once per visit if required, in order to make sure the problem has fully resolved. They will be invited to re-attend for subsequent re-assessment and treatment, at four weekly intervals up to a maximum of three treatments, as informed by the preparatory survey of specialists. In this time, if the assessment manoeuvres are negative, this will end the participants involvement in the study.

Any patients that do not respond to three treatments, will then be referred to the acute vertigo service for review by a Consultant Neurologist, as per the consensus of specialists from the preparatory survey. They will also be invited to have high resolution 3D reconstructive magnetic resonance imaging of the inner ear. The imaging will be compared to a purposive group of BPPV patients (n=12) whose symptoms resolved within three manoeuvres to detect any anatomical difference that may explain the lack of resolution. Differences in the angle, orientation, width, and the viscosity of the endolymph will be compared to responders, using computational

methods. This may then allow adjustments to be made to the treatment sequence to optimally treat and resolve the BPPV.

Participants that do not have the nystagmus characteristics expected for BPPV within the positional tests, or present with abnormalities in the oculomotor examination indicating central problems, will be referred to the acute vertigo service for further assessment and management by a Neurologist. This may include medication or further imaging. This will end the participants involvement in the study.

Any participant will be invited to contact the research team, at any point during the recruitment phase of the study, should they feel their symptoms have recurred. They will then follow the same intervention pathway as new participants (see Figure 1).

Statistical analysis

Descriptive and predictive statistical analyses will be performed. Categorical data will be presented as frequencies with percentages. Continuous data will be presented as means and Standard Deviations (±SD) where data is normally distributed. Medians and Inter-Quartile Ranges (IQR) will be used when the data is not normally distributed. Appropriate regression models will be used to investigate the association between individual clinical features and diagnoses of peripheral or central causes of positional nystagmus/dizziness.

3D reconstructive MRI scans of participants not responding to BPPV manoeuvres, will be compared to those that do respond to explore if there is an anatomical difference. How?

Delphi expert opinion

In addition to this observational study, a modified Delphi technique will be used to gain consensus from a multi disciplinary international group of clinicians working with vestibular conditions (Trevelyan and Robinson, 2015). This will include Consultants with a primary interest and with >5 years experience in clinical neuro-otology or audiovestibular medicine, ENT Consultants, Neurology Consultants or GPs working with dizziness. In addition, physical therapists, and audiologists with >5 year's experience of working with vestibular conditions will also be included. There will be three rounds to the Delphi process.

Participants will be identified through the membership lists of the European Academy of Neurology (EAN) specialist panel of neuro-otologists, Barany Society, the vestibular rehabilitation special interest groups for the American Physical Therapy Association (APTA), the Association of Chartered Physiotherapists Interested in Vestibular Rehabiliation (ACPIVR), and the British Society of Audiology (BSA).

Round one

This will be a survey to explore participants management of the different subtypes of BPPV, expanding the preparatory survey results. The responses will be anonymous and confidential.

Membership secretaries will be sent an email by a member of the research team, with specific wording to forward on to members asking them to participate. Within the email invitation, will be a link for the free survey using Google forms.

The membership secretaries will send a reminder email one week later, in order to maximise participation.

Round two

Round two will be informed by round one. Statements will be developed from the survey answers with the highest scores, and these will be emailed via the membership secretaries of the above mentioned international groups.

Respondents will be asked to rate how much they agree or disagree with the statements using a six-point scale. Choosing a scale without a midpoint will avoid neutral responses and will make respondents make a positive or negative decision about each question (Trevelyan and Robinson, 2015). They will also be asked to add any statements they feel should be included or suggest any that need rewording.

The membership secretaries will send a reminder email one week later. Consensus per statement will be defined when >75% of the responses are in the lower or higher tertile (Boulkedid *et al.*, 2011).

The responses will be anonymous and confidential. However, at this stage, respondents will be asked if they would like to be part of a panel discussion in order to discuss any points that have not reached consensus. If respondents are interested, they will be asked to provide an email address and name so they can be contacted.

Round three

This will involve a panel discussion of volunteers from round two. A time will be established that will accommodate as many of those interested as possible, accounting for time differences and the individuals' commitments.

The panel discussion will attempt to reach a majority consensus on any remaining statements. Any points that do not reach majority consensus, will have recommendations made for future research.

Dissemination, outputs and anticipated impact

The results from this project will be submitted to relevant peer reviewed journals for publication. I anticipate at least three papers that will include describing the nystagmus features of the largest cohort of patients with positional nystagmus and any key distinguishing features, sharing the outcome of the International and multidisciplinary Delphi consensus, and a paper to share the findings from the first study to date undertaking 3D reconstructive imaging of the inner ear.

I will also submit these results to specialist conferences, to present as either a poster or platform presentation. This includes the Barany Society, and the British Society of Neuro-Otology. These conferences are attended by specialist doctors, physiotherapists, audiologists, and other members of the multidisciplinary team working with dizziness and balance patients, and are internationally renowned. ? ED/other medical ones ?

As a member of the physiotherapy interest group (ACPIVR), I will also share my results at either annual national or regional training events, and through the quarterly journal. I will embed the findings in any education I continue to provide through national and international courses, and through university masters programs.

I will also share the results through a range of social media platforms, including the research unit at UCL, ACPIVR, my own personal accounts, and the Ménière's Disease society. I will also

share the findings as a newsletter to the service users that assisted in recruitment, both in the community and at UCLH. I will also offer to present the findings at any local departmental meetings.

As part of this project, I plan to set up a PPI advisory group and will seek their help on how best to disseminate the findings so results are shared with the people it affects, and improves care in the future.

Project management

The project has been discussed with the NIHR Research Support Service (RSS) and xxxxx at the UCL/UCLH Joint Research Office. The project has been developed with support from a specialist Neurologist that leads the SENSE research unit in the Institute of Neurology and a specialist Neuroradiologist.

Costings have been approved by the xxxxx based in xxxxx.

Ethics

Once funding is approved, I will apply for internal ethical approval through the Joint Research Office at UCLH. I will also complete the Integrated Research Application System (IRAS) form and apply for NHS ethical approval, as this study will include patients.

Success criteria